

Difference Between Stop and Wait, GoBackN and Selective Repeat

Flow control in networking manages the rate of data transmission between devices to ensure efficient processing and prevent data overflow. When a device receives more data than it can handle, it risks losing information or needing retransmission. Flow control protocols guide the sender on how much data to send, and typically require an acknowledgment from the receiver before more data is sent.

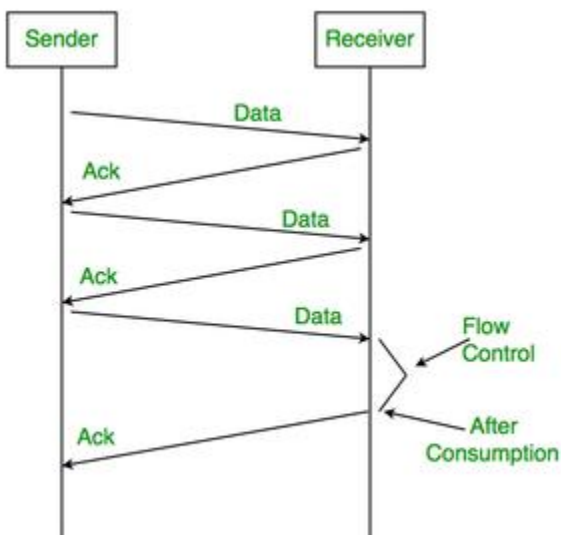
There are three main flow control methods:

- Stop and Wait ARQ,
- Go Back N ARQ
- Selective Repeat ARQ

Reliable data transfers are one of the primary concerns in computer networking. This service department lies in the hands of [TCP](#). Their major flow control protocols are stop and Wait, Go Back N, and Selective Repeat.

Stop and Wait ARQ

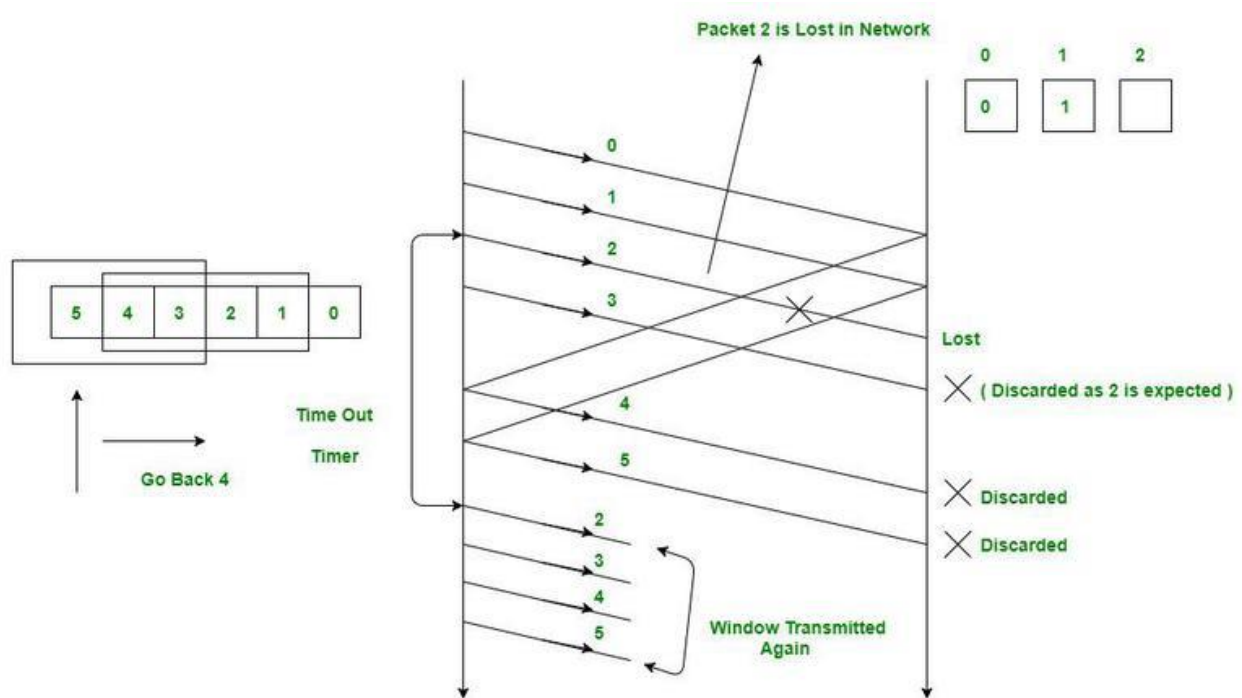
The sender sends the packet and waits for the acknowledgment of the packet. Once the acknowledgment reaches the sender, it transmits the next packet in a row. If the acknowledgment does not reach the sender before the specified time, known as the timeout, the sender sends the same packet again.



Stop and Wait ARQ

Go Back N ARQ

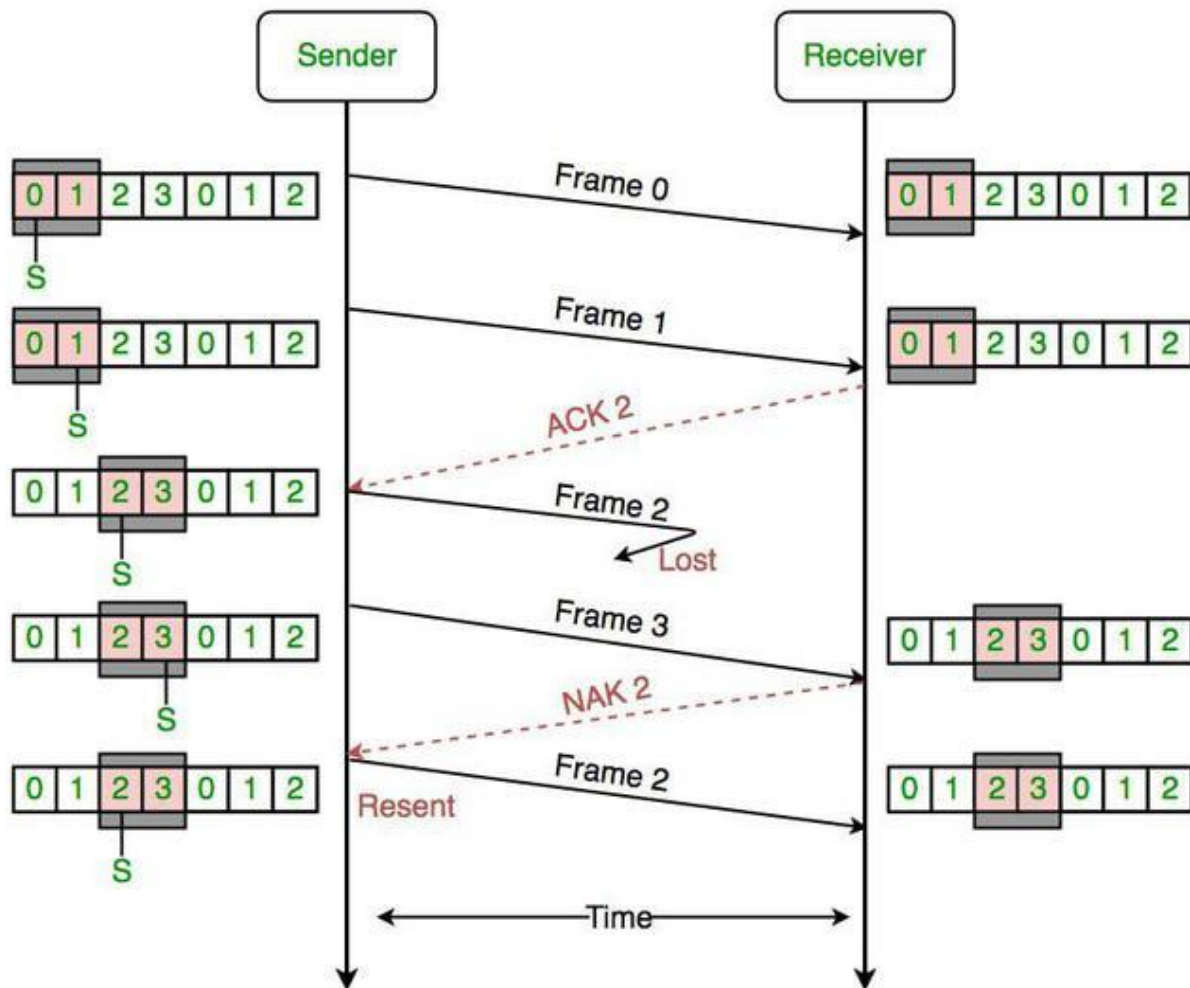
The sender sends N packets which are equal to the window size. Once the entire window is sent, the sender then waits for a cumulative acknowledgement to send more packets. On the receiver end, it receives only in-order packets and discards out-of-order packets. As in case of packet loss, the entire window would be re-transmitted.



Go Back N ARQ

Selective Repeat ARQ

The sender sends packets of window size N and the receiver acknowledges all packets whether they were received in order or not. In this case, the receiver maintains a buffer to contain out-of-order packets and sorts them. The sender selectively re-transmits the lost packet and moves the window forward.



Selective Repeat ARQ

Difference Between Stop and Wait, GoBackN and Selective Repeat

Properties	Stop and Wait ARQ	Go Back N ARQ	Selective Repeat ARQ
Sender window size	1	N	N
Receiver Window size	1	1	N

Properties	Stop and Wait ARQ	Go Back N ARQ	Selective Repeat ARQ
Minimum Sequence number	2	N+1	2N
Efficiency	$1/(1+2*a)$	$N/(1+2*a)$	$N/(1+2*a)$
Type of Acknowledgement	Individual	Cumulative	Individual
Supported order at the Receiving end	In-order delivery only	In-order delivery only	Out-of-order delivery as well
Number of retransmissions in case of packet drop	1	N	1
Transmission Type	Half duplex	Full duplex	Full duplex
Implementation difficulty	Low	Moderate	Complex

Where,

- a = Ratio of Propagation delay and Transmission delay,
- At $N=1$, Go Back N is effectively reduced to Stop and Wait,
- As Go Back N acknowledges the packets cumulatively, it rejects out-of-order packets,
- As Selective Repeat supports receiving out-of-order packets (it sorts the window after receiving the packets), it uses Independent Acknowledgement to acknowledge the packets.

Password Authentication Protocol (PAP)

The Password Authentication Protocol (PAP) is a basic but important method used to check who is accessing network services, especially in systems that use the Point-to-Point Protocol (PPP). PAP is easy to use because it simply verifies user credentials to allow network access. This simplicity comes with security risks since it sends passwords in a form that can be easily seen and stolen.

This article explains how PAP works, its features, and why it's still used today. We will also look at its security risks and discuss safer alternatives to protect sensitive information.

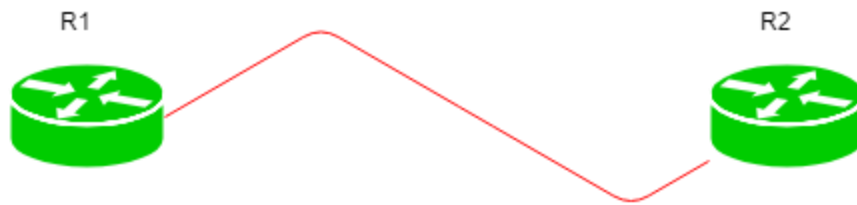
Understanding Password Authentication Protocol (PAP)

The Password Authentication Protocol (PAP) operates primarily in network environments where a simple authentication mechanism is sufficient. Typically used in conjunction with the Point-to-Point Protocol (PPP), PAP starts the authentication process when a client attempts to establish a connection with a server. During this initiation phase, the client sends a request to the server indicating its desire to authenticate its identity. This straightforward handshake sets the stage for the transmission of authentication details.

Once the initial request is made, PAP proceeds to the authentication phase where the client transmits its username and password directly to the server. These credentials are sent as plaintext, meaning they are not encrypted or hidden in any manner. This exposes them to potential interception by unauthorized parties who may be monitoring the network traffic. The server then receives these credentials and begins the process of verifying them against its stored records to determine if the access should be granted.

If the credentials provided by the client match those on the server then the access is granted, and the client is notified of the successful authentication. Conversely, the server rejects the authentication request if there is a mismatch. In some implementations the server might allow the client multiple attempts to enter the correct credentials, which could potentially open up [vulnerabilities](#) such as brute force attacks. This lack of robust security measures underscores the simplicity but also highlights the significant risks associated with using PAP in environments where security is a concern.

Configuration Steps for PAP



PAP EXAMPLE

There is a small topology in which there are 2 routers namely R1 and R2. R1 has IP address 10.1.1.1/30 on s0/0 and R2 has IP address 10.1.1.2/30 on s0/0.

First, we will create a local database on R1 by providing a username and password:

```
R1(config)#username Router1 password GeeksforGeeks
```

Configuring local database on R2:

```
R2(config)#username Router2 password GeeksforGeeks
```

Remember, by default HDLC is configured on Cisco routers therefore first we have to change the encapsulation to PPP and enable PAP.

```
R1(config)# int s0/0
```

```
R1(config-if)#encapsulation ppp
```

```
R1(config-if)#ppp authentication pap
```

```
R1(config-if)#ppp pap sent-username Router2 password GeeksforGeeks
```

Enabling PAP on R2:

```
R2(config)# int s0/0
```

```
R2(config-if)#encapsulation ppp
```

```
R2(config-if)#ppp authentication pap
```

```
R2(config-if)#ppp pap sent-username Router1 password GeeksforGeeks
```

Here, notice that username and password are case-sensitive. Also, on router R1 we have to give a username and password.

Note: This command can also be used on the router which wants to authenticate (calling [router](#)) in case of one-way authentication i.e only the calling router will authenticate. If two-way authentication, i.e both client and remote device are going to authenticate to each other, is operating then we have to make a local database and use this command on both devices.

In addition, if we want to use CHAP first and PAP as a backup when CHAP fails, we can configure it by the command.

```
R1(config)#int s0/0
```

```
R2(config-if)#ppp authentication chap pap
```

Also if we want CHAP as a backup then use the command.

```
R1(config)#int s0/0
```

```
R2(config-if)#ppp authentication pap chap
```

Where PAP is Used?

Password Authentication Protocol (PAP) is used in several contexts where simple, password-based authentication is sufficient, particularly in environments where high security is not the primary concern. Here's where PAP is commonly used:

- **Dial-Up Networks:** PAP is often used in dial-up networking where a user connects to the internet through a modem. In these scenarios, PAP facilitates the authentication of the user's credentials (username and password) against the [Internet Service Provider's](#) (ISP) systems. This usage has declined with the decrease in dial-up connections in favor of broadband and fiber connections.
- **VPN Connections:** [Virtual Private Networks](#) (VPNs) sometimes use PAP, especially in legacy systems where other more secure options are not available or compatible. PAP allows remote users to authenticate their credentials when accessing a network through a [VPN tunnel](#).
- **Point-to-Point Protocol (PPP) Links:** In network configurations that use PPP, such as connections between two routers over serial links, PAP can be employed for authentication. PPP supports multiple authentication methods, and PAP is the simplest, used particularly where both endpoints are controlled, reducing the risk of credential interception.
- **Network Access Servers (NAS):** Network Access Servers, which provide access services to a larger network or the [internet](#), might use PAP for authenticating connecting clients. This is common in smaller or less security-sensitive environments.
- **Legacy Systems and Backward Compatibility:** In older systems or networks where newer protocols are not supported due to hardware or software limitations, PAP remains a viable authentication option. It provides a basic level of security without the need for updates or changes to existing infrastructure.

Difference Between PAP and CHAP

PAP (Password Authentication Protocol)	CHAP (Challenge Handshake Authentication Protocol)
Uses Two-way handshake for authentication, sends username and password in plaintext.	Uses Three-way handshake for authentication, uses a challenge-response mechanism where only the hash of the password is sent.
Offers minimal security as the passwords are transmitted in plaintext, making it less secure.	Provides enhanced security compared to PAP, does not transmit passwords directly, uses hashed values, and periodic challenges.
Highly vulnerable to eavesdropping and replay attacks.	Less vulnerable to replay attacks due to the random challenge used in each authentication session.
Suitable for low-security environments or where legacy systems require simple authentication.	Preferred in environments where enhanced security is needed but still manageable with password-based authentication.
Simple to implement with minimal resource requirements.	More complex due to cryptographic requirements and management of challenges, requires more processing power.

Advantages of PAP over CHAP

- **Ease of Implementation:** PAP is simpler to implement and configure compared to CHAP, making it beneficial for environments where advanced security is not a priority.
- **Compatibility:** Due to its basic nature, PAP offers broad compatibility with older systems and devices that may not support the [cryptographic functions](#) required by CHAP.
- **Lower Resource Usage:** PAP does not involve cryptographic calculations, which reduces its processing requirements, making it suitable for older hardware or embedded systems with limited resources.

- **Administrative Simplicity:** PAP allows for easier troubleshooting and monitoring due to its straightforward mechanism, providing direct and immediate feedback on authentication failures.